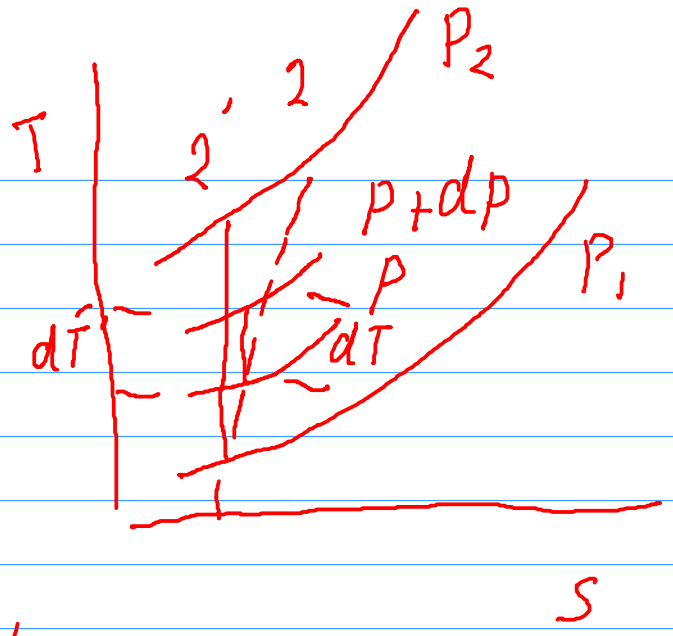
$$\eta_{pc} = \frac{dT'}{dT}$$



$$pv^n = \text{const}$$

$$T_2'/T_1 = (P_2/P_1)^{\gamma-1/\gamma}$$

$$T_2/T_1 = (P_2/P_1)^{\frac{n-1}{n}}$$

$$\eta_{pc} = \frac{dT' + T = T}{dT} = \frac{\frac{dT'+T}{T} - 1}{dT/T}$$

$$\eta_{pc} \cdot \frac{dT}{T} = \left(\frac{P+dP}{P}\right)^{\gamma-1/\gamma} - 1$$

$$\left(1 + \frac{dP}{P}\right)^{\gamma-1/\gamma}$$

$$\eta_{pc} \cdot \frac{dT}{T} = 1 + \frac{\gamma-1}{\gamma}\frac{dP}{P} - 1$$

$$dT/T = \frac{\gamma-1}{\gamma \eta_{pc}} \cdot dP/P$$

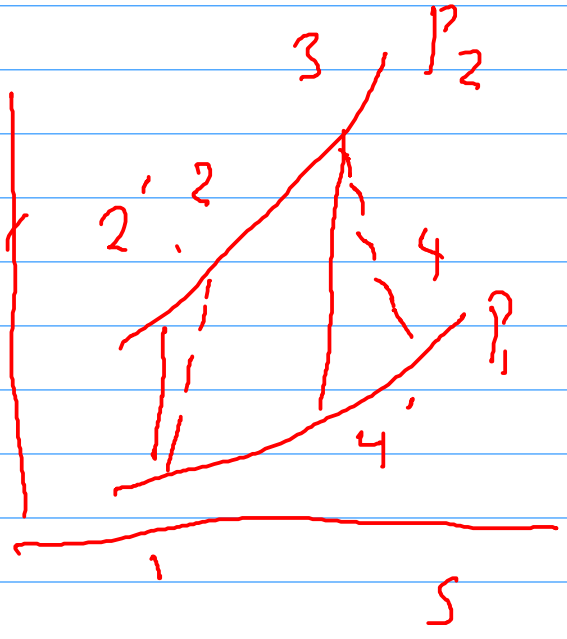
$$\text{If, } T_2/T_1 = (P_2/P_1)^{\gamma-1/\gamma \cdot \eta_{pc}}$$

$$\text{If } T_3/T_4 = (P_3/P_4)^{\frac{\gamma-1}{\gamma} \cdot \eta_{pe}}$$

$$\frac{\gamma-1}{\gamma \eta_{pc}} = \frac{\eta-1}{\eta}$$

$$T_2 = T_1 (P_2/P_1)^{\frac{\gamma-1}{\gamma \eta_{pc}}}$$

$$T_3/T_4 = (P_3/P_4)^{\frac{\gamma-1}{\gamma} \cdot \eta_{pe}}$$



$$\dot{W}_c = \dot{m} C_{p_0} (T_2 - T_1)$$

$$\dot{W}_T = \dot{m} C_{p_0} (T_3 - T_4)$$

$$\dot{Q} = \dot{m} C_{p_{0\,avg}} (T_3 - T_2)$$

$$P = \dot{W}_T - \dot{W}_c$$

$$\eta = \frac{P}{\dot{Q}}$$

$$\eta_{isc} = \frac{T_{2'} - T_1}{T_2 - T_1} = \frac{(r_p)^{\frac{\gamma-1}{\gamma}} - 1}{(r_p)^{\frac{\gamma-1}{\gamma}\eta_c} - 1}$$

$$\eta_{isc} = \frac{18^{0.286} - 1}{18^{0.286/0.82} - 1} =$$

$$\eta_{isc} = 0.738$$